Geophysical Journal International

Geophys. J. Int. (2018) **212,** 390–399
Advance Access publication 2017 October 10
GJI Seismology

doi: 10.1093/gji/ggx433

Calculation of differential seismograms using analytic partial derivatives—II Regional waveforms

Shaoqian Hu* and Lupei Zhu

Department of Earth and Atmospheric Sciences, Saint Louis University, St Louis, MO 63103, USA. E-mail: zhul@slu.edu

Accepted 2017 October 7. Received 2017 October 5; in original form 2017 May 11

SUMMARY

Solving seismic waveform inverse problem by the linearization approach requires calculating partial derivatives of seismograms (differential seismograms) with respect to velocity model parameters. Here we extend our previous work on calculation of differential teleseismic receiver functions using analytic partial derivatives to regional waveforms. We derived analytic partial derivatives of displacement kernels for both *P–SV* and *SH* waves generated by a point source in a multilayered half-space. They are expressed explicitly in terms of the analytic partial derivatives of the Haskell propagator matrix and the source vector. Differential seismograms are then computed using the wavenumber–frequency double integration method. We implemented an efficient algorithm to speed up the computation by storing intermediate matrix products. The total computation time for computing differential seismograms with respect to all the layers is only about one extra forward-problem computation time. Numerical tests show that our derivations are correct and the implementation is efficient. Synthetic and real data waveform inversions using more accurate analytic differential seismograms perform better than using finite-difference differential seismograms.

Key words: Waveform inversion; Theoretical seismology; Wave propagation.

1 INTRODUCTION

Full waveform inversion (FWI) has become an increasingly important tool to determine sub-surface structure in both active and passive-source seismology recently (e.g. Virieux & Operto 2009; Liu & Gu 2012). Since seismic waveforms contain more information about Earth structure than arrival times of individual phases, FWI is believed to be able to reveal more detailed Earth structure with less non-uniqueness than the traditional travel-time tomography. Powered by fast high-performance computers and accumulation of high-quality waveform records, 1-D and 3-D FWI's have been successfully applied in various areas ranging from local to global scales (e.g. Chen *et al.* 2007; Tape *et al.* 2009; Fichtner *et al.* 2009; Zhu *et al.* 2012; Bozdağ *et al.* 2016).

Waveform inversion for earth structure is intrinsically nonlinear. Though nonlinear inversion methods are used occasionally (e.g. Sambridge & Drijkoningen 1992; Hrubcová *et al.* 2016), iterative linearization approaches are often adopted for efficiency (e.g. Holt & Wallace 1990; Kormendi & Dietrich 1991; Xu & Wiens 1997; Gao *et al.* 2006; Wang *et al.* 2014). During each iteration, partial derivatives of waveforms (also called differential seismograms) with respect to model parameters are used to linearize the inverse problem. Computation of differential seismograms is usually the main computing-time burden in the whole inversion process and their accuracy can be critical in determining the convergence of inversion.

For FWI's using local and regional seismic waveforms, crust and upper mantle structure can often be approximated as 1-D layered models (e.g. Xu & Wiens 1997; Gao *et al.* 2006; Brandt *et al.* 2012; Hrubcová *et al.* 2016). Even for 2-D and 3-D seismic data collected in geologically complicated regions it is feasible to apply FWI using local 1-D model to the Common-Midpoint-Point gathers or in the τ-p domain (e.g. Mallick & Adhikari 2015; Pafeng *et al.* 2017). In all these cases, fast algorithms for computing theoretical seismograms, such as the reflectivity method (Fuchs & Müller 1971; Kennett 1983) and frequency-wavenumber (FK) integration method (Wang & Herrmann 1980; Zhu & Rivera 2002) are available. However, the differential seismograms are mostly computed numerically using the finite-difference (FD) method (e.g. Randall 1994; Sen & Roy 2003). Dietrich & Kormendi (1990) gave explicit analytic derivatives of *P–SV* plane waves for a thin layer in a vertically heterogeneous medium by using the first-order perturbation analysis. Numerical tests showed that the accuracy deteriorates with increase in incidence angle and layer thickness (Moinet & Dietrich 1998). Zeng & Anderson (1995) also used the perturbation analysis and derived a method to compute differential seismograms for a finite-thickness layer in a layered half-space. Du *et al.* (1998) presented an analytic method to compute differential seismograms based on the *P–SV*-wave multimode summation formalism. It works only for relatively long-period surface waves.

* Now at: Laboratory of Seismology and Physics of Earth's Interior, School of Earth and Space Sciences, University of Science of Technology of China, Hefei, Anhui 230026, China.

Recently, we derived explicit analytic partial derivatives of teleseismic receiver functions based on the Haskell propagate matrix formalism (Hu & Zhu 2017). In this paper, we extend the previous work to local and regional waveforms by adding a point source in the multilayered model. Explicit analytic partial derivatives of displacement kernels of both *P–SV* and *SH* waves generated by the point source will be obtained in terms of analytic partial derivatives of the source terms, compound matrices of *P–SV* Haskell matrices, and *SH* Haskell matrices. Differential seismograms can then be calculated using the frequency-wavenumber double integration method. We will use numerical tests to verify our method's correctness and efficiency and apply it to waveform inversion to demonstrate its advantages over the FD differential seismograms. We will also compare our analytic derivatives with those based on the perturbation analysis with Born approximation. We show that the two are equivalent when the perturbation to the source layer is handled correctly.

2 THEORY

2.1 Displacement solutions from a point source in a multilayered medium

For a point source in a multilayered model as shown in Fig. 1, the surface displacement solution has been derived using the Thomson–Haskell propagator matrix method (Haskell 1953, 1963). Here we follow the derivations in Zhu & Rivera (2002). The displacement-stress vector $\mathbf{b}(z)$ in each layer satisfies a 1st-order linear ODE:

$$\frac{\mathrm{d}\mathbf{b}(z)}{\mathrm{d}z} = \mathbf{M}\mathbf{b}(z), \tag{1}$$

where $\mathbf{M}$ is a z-independent medium matrix. Its general solution can be expressed as

$$\mathbf{b}(z) = \mathbf{E}\boldsymbol{\Lambda}(z)\mathbf{w}, \tag{2}$$

where $\mathbf{w}$ is a constant vector of wave amplitude in the layer. The displacement-stress vectors at the bottom and top of the layer are connected by

$$\mathbf{b}(z_n) = \mathbf{a}_n\mathbf{b}(z_{n-1}), \tag{3}$$

where

$$\mathbf{a}_n = \mathbf{E}\boldsymbol{\Lambda}(z_n - z_{n-1})\mathbf{E}^{-1}, \tag{4}$$

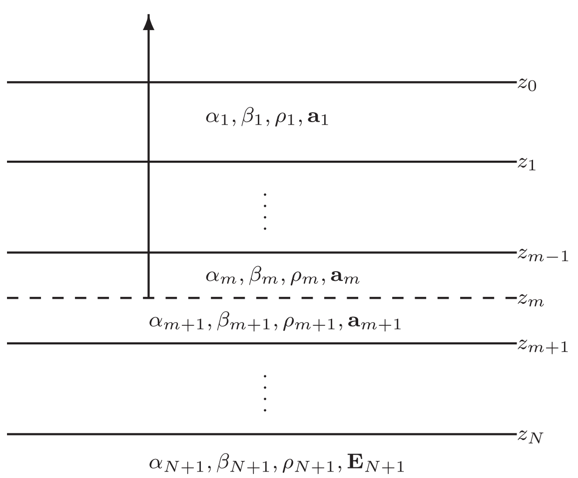

Figure 1. A model of N uniform layers over a half-space. The source is located in at a virtual interface between layers m and $m + 1$.

is called the Thomson–Haskell propagator matrix. Components of $\mathbf{a}$, $\mathbf{E}$, $\boldsymbol{\Lambda}$, and $\mathbf{E}^{-1}$ are given in Zhu & Rivera (2002).

$\mathbf{b}(z)$ is continuous everywhere except at the source depth,

$$\mathbf{b}(z_m^+) - \mathbf{b}(z_m^-) = \mathbf{s}, \tag{5}$$

where $\mathbf{s}$ is the displacement-stress jump caused by the source. The surface displacements are found by using a traction-free condition at the surface and the radiation condition in the bottom half-space. For the *P–SV* wave, the displacement kernels are

$$\begin{bmatrix} U_r \\ U_z \end{bmatrix} = \frac{1}{F_R}\begin{bmatrix} R_{22} & -R_{12} \\ -R_{21} & R_{11} \end{bmatrix}\begin{bmatrix} X_{1i}s_i \\ X_{2i}s_i \end{bmatrix}, \tag{6}$$

where

$$\mathbf{X} = \mathbf{E}_{N+1}^{-1}\,\mathbf{a}_N\,\cdots\,\mathbf{a}_{m+1}, \tag{7}$$

$$\mathbf{R} = \mathbf{E}_{N+1}^{-1}\,\mathbf{a}_N\,\cdots\,\mathbf{a}_1, \tag{8}$$

$$F_R = R_{11}R_{22} - R_{12}R_{21}. \tag{9}$$

F_R is called the Rayleigh denominator.

Computation using eq. (6) suffers from loss of significant digits and numerical errors due to exponential terms in the propagator matrices. A solution is to use the compound matrix instead (e.g. Dunkin 1965; Watson 1970; Wang & Herrmann 1980),

$$\mathbf{a}|_{kl}^{ij} = a_{ik}a_{jl} - a_{il}a_{jk}. \tag{10}$$

Eq. (6) is replaced by,

$$U_k = \frac{s_i\mathbf{X}|_{ij}^{12}Z_{jk}}{F_R}, \tag{11}$$

where $U_1 = -U_z$, $U_2 = U_r$,

$$\mathbf{Z} = \mathbf{a}_m\cdots\mathbf{a}_1, \tag{12}$$

$$\mathbf{X}|_{ij}^{12} = (\mathbf{E}_{N+1}^{-1})|_{mn}^{12}\mathbf{a}_N|_{op}^{mn}\cdots\mathbf{a}_{m+1}|_{ij}^{uv}, \tag{13}$$

$$F_R = (\mathbf{E}_{N+1}^{-1})|_{mn}^{12}\mathbf{a}_N|_{op}^{mn}\cdots\mathbf{a}_1|_{12}^{uv}. \tag{14}$$

For the *SH* system, the displacement kernel is

$$U_\theta = \frac{X_{1i}s_i}{F_L}, \tag{15}$$

where $F_L = R_{11}$ is called the Love denominator. $\mathbf{X}$ and $\mathbf{R}$ take the same forms as in eqs (7) and (8) except using the 2×2 *SH* Haskell matrices.

2.2 Analytic partial derivatives of displacement kernels

Let x_l represent an independent model parameter, for example, α, β, ρ of layer l. Partial derivatives of the *P–SV* displacement kernels with respect to x_l can be obtained by applying the quotient rule to eq. (11), depending on the location of layer l relative to the source layer:

(i) $l < m$,

$$\frac{\partial U_k}{\partial x_l} \equiv U_{k,l} = \frac{s_i\mathbf{X}|_{ij}^{12}Z_{jk,l}}{F_R} - \frac{F_{R,l}}{F_R}U_k, \tag{16}$$

(ii) $l = m$ or $l = m + 1$,

$$U_{k,l} = \frac{s_i\left(\mathbf{X}|_{ij,m+1}^{12}Z_{jk} + \mathbf{X}|_{ij}^{12}Z_{jk,m}\right)}{F_R} + \frac{s_{i,l}\mathbf{X}|_{ij}^{12}Z_{jk}}{F_R} - \frac{F_{R,m+1} + F_{R,m}}{F_R}U_k, \tag{17}$$

(iii) $l > m + 1$,

$$U_{k,l} = \frac{s_i \mathbf{X}|^{12}_{ij,l} Z_{jk}}{F_R} - \frac{F_{R,l}}{F_R} U_k, \tag{18}$$

where

$$Z_{jk,l} = a^{(m)}_{jp} \cdots a^{(l)}_{qr,l} \cdots a^{(1)}_{sk}, \tag{19}$$

$$\mathbf{X}|_{ij,l\le N} = \left(\mathbf{E}^{-1}_{N+1}\right)|^{12}_{mn} \mathbf{a}_N|^{mn}_{op} \cdots \mathbf{a}_l|^{qr}_{st,l} \cdots \mathbf{a}_{m+1}|^{uv}_{ij}, \tag{20}$$

$$\mathbf{X}|_{ij,N+1} = \left(\mathbf{E}^{-1}_{N+1}\right)|^{12}_{mn,N+1} \mathbf{a}_N|^{mn}_{op} \cdots \mathbf{a}_{m+1}|^{uv}_{ij}, \tag{21}$$

$$F_{R,l\le N} = \left(\mathbf{E}^{-1}_{N+1}\right)|^{12}_{mn} \mathbf{a}_N|^{mn}_{op} \cdots \mathbf{a}_l|^{qr}_{st,l} \cdots \mathbf{a}_1|^{uv}_{12}, \tag{22}$$

$$F_{R,N+1} = \left(\mathbf{E}^{-1}_{N+1}\right)|^{12}_{mn,N+1} \mathbf{a}_N|^{mn}_{op} \cdots \mathbf{a}_1|^{uv}_{12}. \tag{23}$$

Analytic partial derivatives of $\mathbf{a}$ and $\mathbf{E}^{-1}_{N+1}$ have been given in Hu & Zhu (2017). We give the partial derivatives of their compound matrices and $\mathbf{s}$ in Appendix A.

Analytic partial derivatives of *SH* displacement kernel are:

(i) $l < m$,

$$U_{\theta,l} = -\frac{F_{L,l}}{F_L} U_\theta, \tag{24}$$

(ii) $l = m$ or $l = m + 1$,

$$U_{\theta,l} = \frac{X_{1i,m+1} s_i}{F_L} + \frac{X_{1i} s_{i,l}}{F_L} - \frac{F_{L,m+1} + F_{L,m}}{F_L} U_\theta, \tag{25}$$

(iii) $l > m + 1$,

$$U_{\theta,l} = \frac{X_{1i,l} s_i}{F_L} - \frac{F_{L,l}}{F_L} U_\theta, \tag{26}$$

where

$$X_{1i,l\le N} = \left(\mathbf{E}^{-1}_{N+1}\right)_{1p} a^{(N)}_{pq} \cdots a^{(l)}_{rs,l} \cdots a^{(m+1)}_{ti}, \tag{27}$$

$$X_{1i,N+1} = \left(\mathbf{E}^{-1}_{N+1}\right)_{1p,N+1} a^{(N)}_{pq} \cdots a^{(m+1)}_{ti}, \tag{28}$$

$$F_{L,l\le N} = \left(\mathbf{E}^{-1}_{N+1}\right)_{1p} a^{(N)}_{pq} \cdots a^{(l)}_{rs,l} \cdots a^{(1)}_{t1}, \tag{29}$$

$$F_{L,N+1} = \left(\mathbf{E}^{-1}_{N+1}\right)_{1p,N+1} a^{(N)}_{pq} \cdots a^{(1)}_{t1}. \tag{30}$$

The analytic partial derivatives of the *SH* Haskell matrices and source vectors are given in Appendix B.

3 IMPLEMENTATION AND TESTS

It is clear from the above analytic results that computing analytic differential seismograms with respect to one layer parameter requires calculating the partial derivative of Haskell matrix (or its compound matrix) of each layer and multiplying it with Haskell (or compound) matrices of all other layers. So the total computation time of all differential seismograms for an N-layer model is proportional to N^2, that is, the computation complexity is $O(N^2)$. Randall (1994) proposed an optimization algorithm to reduce computation complexity to $O(N)$ by storing the products of Haskell matrices so that they do not need to be repeatedly calculated. We implemented this optimization by using two separate loops. In the first loop from the bottom to the top of the model, the Haskell matrix of each layer and its product with all the Haskell matrices of layers beneath it are calculated and saved. In the second loop from the top to the bottom, the partial derivatives of each layer's Haskell matrix are calculated. The partial derivatives of the displacement kernels with respect to the layer are then calculated using the stored matrix products. Numerical tests show the computation time of our implementation is linearly proportional to N and is about one extra forward-problem computation time (see Hu & Zhu 2017).

Table 1. The Hadley–Kanamori velocity model.

Th. (km)	β (km s^{-1})	α (km s^{-1})	ρ (g cm^{-3})	Q_β	Q_α
5.5	3.18	5.50	2.53	600	1200
10.5	3.64	6.30	2.79	600	1200
16.0	3.87	6.70	2.91	600	1200
	4.50	7.80	3.27	900	1800

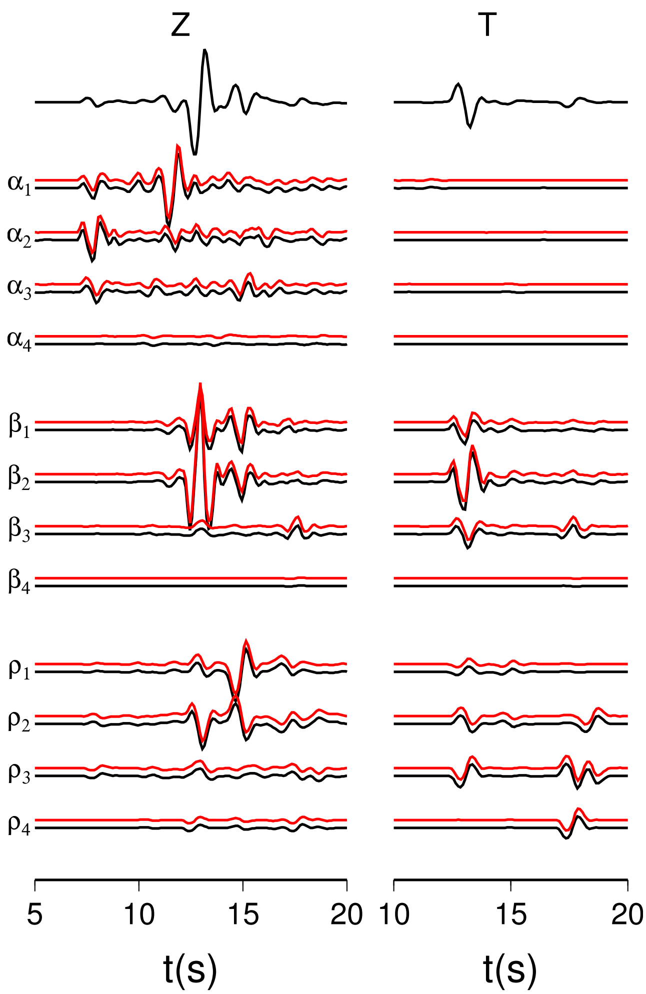

Figure 2. Comparison of vertical (left) and transverse (right) differential seismograms with respect to velocities and density of each layer using FD (black) and analytic partial derivatives (red). Amplitudes of differential seismograms with respect to β are scaled down by a factor of 10 for display.

To verify the correctness of our analytic partial derivatives and implementation, we calculated differential seismograms by using the analytic and FD methods using a 1-D southern California velocity model by Hadley & Kanamori (1977) (Table 1). A small perturbation of 0.1 per cent of model parameters was used in the FD computation. The earthquake magnitude is M_w 5.5 at a depth

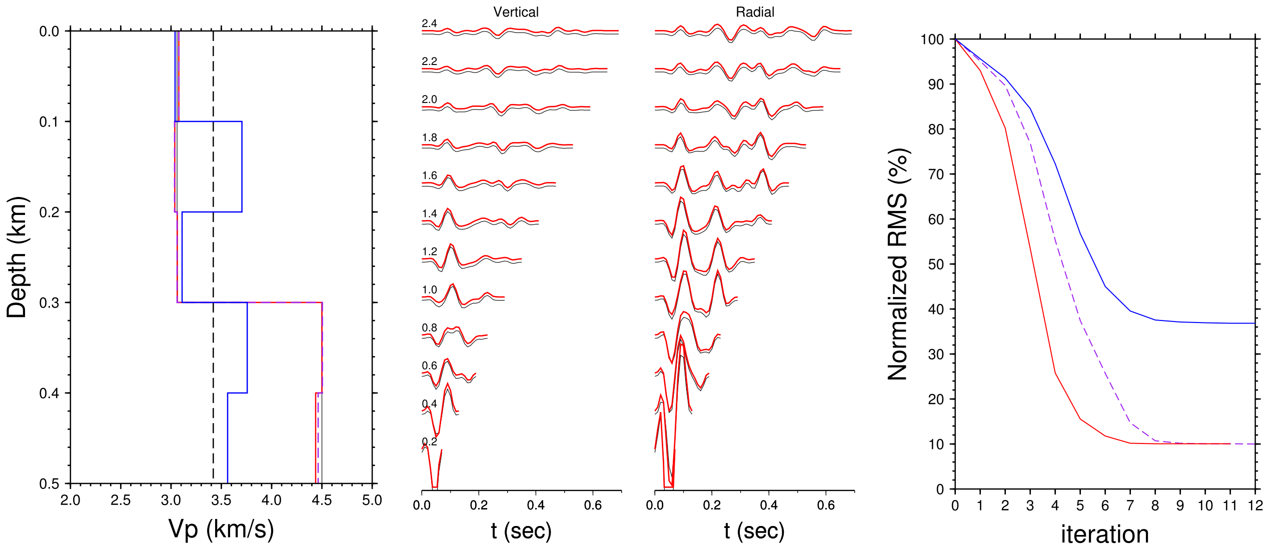

Figure 3. Results of iterative damped-least-squares waveform inversion of synthetic data. Left: the true model (solid black), initial model (dashed black), final model using analytic differential seismograms (red), final model using 1.2 per cent-FD differential seismograms (dashed purple) and final model using 1.5 per cent-FD differential seismograms (blue). Middle: waveform fits of 'observed' seismograms (black) and prediction (red) of the final model using analytic differential seismograms. Epicentre distances in km are labelled next to each trace. Right: decrease of waveform misfit with iteration using the analytic partial derivatives (red), 1.2 per cent-FD (dashed purple), and 1.5 per cent-FD (blue).

of 15 km. The fault plane solution is strike 355°, dip 80° and rake −70°. The station's epicentre distance is 40 km. A band-pass filter of 0.02–2 Hz was applied to the synthetics and differential seismograms. As shown in Fig. 2, the differential seismograms computed by the FD and analytic methods agree well with each other.

We tested using FD and analytic differential seismograms in the waveform inversion with an iterative damped least-squares method. Synthetics data were generated using an explosion source at the surface recorded by stations between 200 and 2400 m in epicentral distance. The velocity model is a 300-m-thick low-velocity layer over a half-space (Fig. 3). We started with a simple uniform half-space model. *P*-wave velocities were determined with a fixed α/β ratio of 1.8 and the empirical relationship $\rho = 0.77 + 0.32\alpha$ between density and α (Berteussen 1977; Owens *et al.* 1984). Finite perturbations of 1.2 per cent and 1.5 per cent were used in the FD differential seismograms calculation. The damped least-square inversion using the 1.5 per cent-FD step size ended in a local minimum and failed to recover the input model (Fig. 3). Inversions of both the 1.2 per cent-FD and analytic differentiation seismograms were able to recover the input model and to produce good waveform fits to the 'observed' seismograms, but the inversion using analytic derivatives converges more rapidly than the FD inversion (Fig. 3).

We also applied the above inversion technique to waveform data of the April 18, 2008 M_w 5.2 Mt. Carmel, IL earthquake. The focal depth (15 km) and mechanism (strike 114°, dip 90° and rake −4°) of this earthquake were determined by Herrmann *et al.* (2008) and Yang *et al.* (2009). Fig. 4 (top) shows the locations of the event and the stations used. The waveforms were corrected for instrument responses and low-pass filtered at 0.17 Hz using a Gaussian filter. A trapezoid of 3 s in duration was used for the source time function. We started with the central US model (CUS; Wang & Herrmann 1980). To compute FD partial derivatives, we tested two perturbations sizes, 1 per cent and 3 per cent. Iterative inversions using the analytic and 1 per cent-FD partial derivative reduced waveform misfits to the same level and obtained similar final models, but the former converges faster. Iterative inversion using the less accurate FD partial derivative of 3 per cent perturbation ended in a local minimum with about 2 per cent higher RMS misfit (Fig. 4, bottom-right). The final models improve waveform fits notably, both in terms of timing and waveform shape. Compared with CUS, the newly derived models have a lower crust of higher velocity and a deeper Moho as a transitional zone between 55 and 65 km. Similar crustal properties are found from inversion of teleseismic Rayleigh wave dispersion (Chen *et al.* 2016).

4 DISCUSSION

We can compare our explicit partial derivatives of displacement kernels with those derived based on the perturbation analysis. Using the Born approximation, the differential wavefield due to a small perturbation of the model can be shown to be produced by virtual sources whose magnitude is the medium perturbation multiplied by the unperturbed wavefield (e.g. Tarantola 1984; Pan *et al.* 1988). For a multilayered half-space, and assuming that the perturbation layer l is not the source layer, we can show that the corresponding virtual source vector (see Appendix C) is,

$$\mathbf{s}^{\mathrm{v}} = -\int_{z_{l-1}}^{z_l} \mathbf{a}(z_l - z)\mathbf{M}_{l,l}\mathbf{a}(z - z_{l-1})\mathrm{d}z\,\mathbf{b}(z_{l-1}). \tag{31}$$

Zeng & Anderson (1995) eliminated the need to integrate over the layer by taking the perturbation directly to the general displacement-stress solution in eq. (2). They derived a virtual source vector in the form of a wave amplitude jump (Eq. (13) in Zeng & Anderson 1995). The corresponding displacement-stress source vector is (see Appendix C),

$$\mathbf{s}^{\mathrm{v}} = -\mathbf{a}_{l,l}\mathbf{b}(z_{l-1}). \tag{32}$$

Partial derivatives of displacement kernels can then be obtained by substituting $\mathbf{s}^{\mathrm{v}}$ for $\mathbf{s}$ in eqs (11) and (15). For example, using

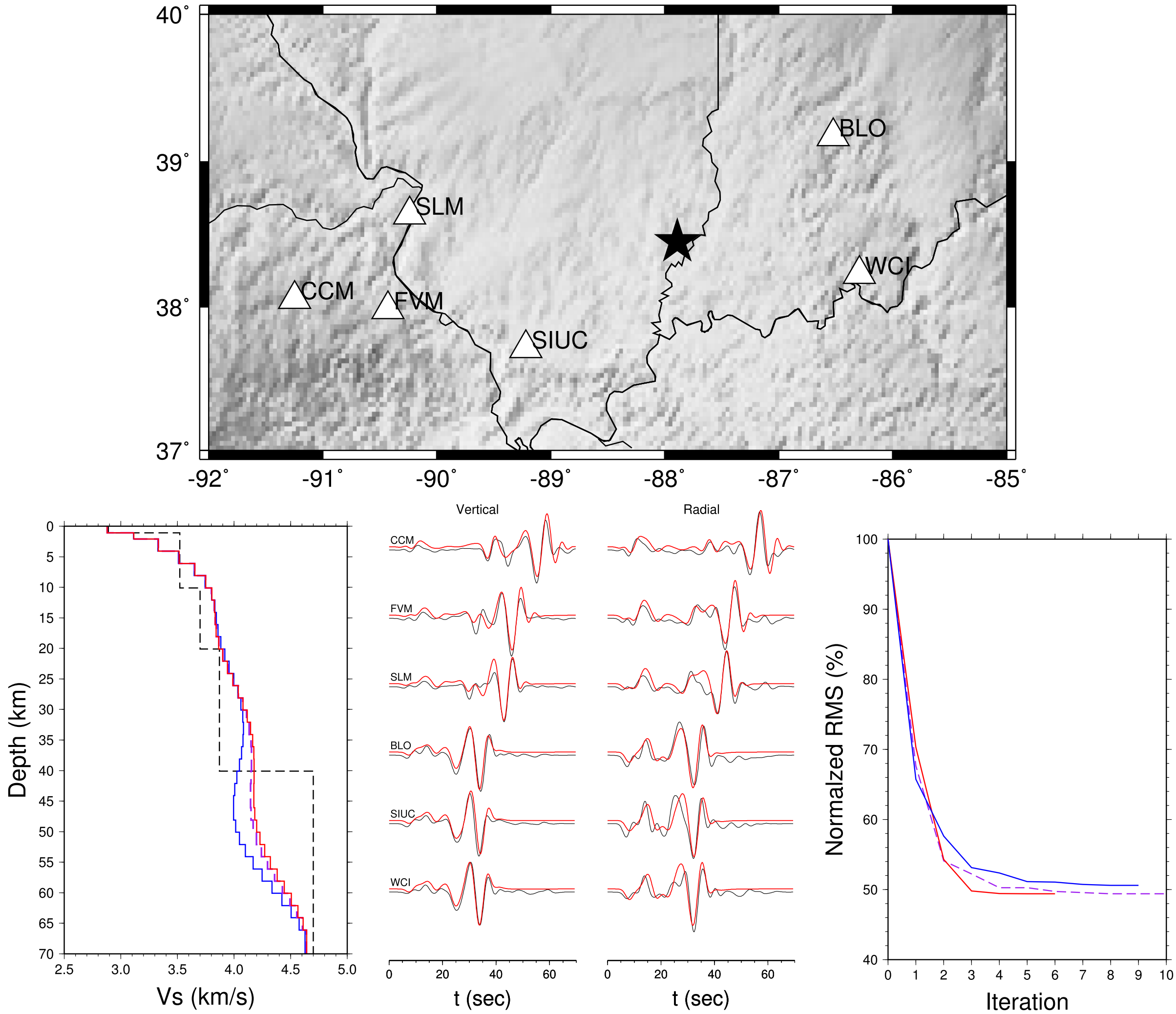

Figure 4. Results of iterative damped-least-squares waveform inversion of real data. Top: locations of the 2008 Mt. Carmel, IL earthquake (star), and distribution of stations (triangles) labelled with station names. Bottom-left: the initial model (dashed black), and obtained models using analytic derivatives (red) as well as FD of 1 per cent (dashed purple) and 3 per cent perturbation (blue). Bottom-middle: waveform fits of observed seismograms (black) and prediction (red) of the final model using analytic differential seismograms. Station names are labelled next to each trace. Bottom-right: decrease of waveform misfit with iteration using the analytic partial derivatives (red), 1 per cent-FD perturbation (dashed purple), and 3 per cent-FD perturbation (blue).

eq. (15), the partial derivatives of the *SH* displacement kernel for $l < m$ are,

$$U_{\theta,l} = \frac{\left(\mathbf{E}_{N+1}^{-1}\right)_{1p} a_{pq}^{(N)} \cdots \left(-a_{rs,l}^{(l)}\right) b_s(z_{l-1})}{F_L}$$
$$= -\frac{\left(\mathbf{E}_{N+1}^{-1}\right)_{1p} a_{pq}^{(N)} \cdots a_{rs,l}^{(l)} \cdots a_{t1}^{(1)}}{F_L} U_\theta, \quad (33)$$

which is the same as in eq. (24). Similarly we can verify the *SH* results in eq. (26) for perturbation beneath the source layer, as well as the *P–SV* results in eqs (16) and (18).

If the source is in the perturbation layer, that is, $l = m$ or $m + 1$, The virtual source vector is found to be (see Appendix C),

$$\mathbf{s}^{\mathrm{v}} = -(\mathbf{a}_{m+1}\mathbf{a}_m)_{,l}\mathbf{b}(z_{m-1}) + \mathbf{a}_{m+1,l}\mathbf{s} + \mathbf{a}_{m+1}\mathbf{s}_{,l}. \quad (34)$$

Comparing the above with eq. (32), we see that the perturbation to the source layer adds two additional terms to the virtual source vector: one from the source vector $\mathbf{s}$ and the other from the perturbation of the source vector $\mathbf{s}_{,l}$. By substituting $\mathbf{s}^{\mathrm{v}}$ above for $\mathbf{s}$ in eqs (11) and (15), it can be shown that eq. (34) gives exactly the three RHS terms of partial derivatives of displacement kernels in eqs (17) and (25). We noted that only the perturbation of the source vector was considered in the virtual source vector of Zeng & Anderson (1995,page 302) but the contribution from the source vector $\mathbf{s}$ was overlooked.

These comparisons confirm that the perturbation analysis with the Born approximation can be used to derive exact analytic partial derivatives of displacement kernels in a multilayered half-space. The Born approximation only requires the perturbation to be small, which is always satisfied by the infinitesimal change of model parameter in partial derivatives. There is no limit on the thickness of the perturbation layer because the combined effect of the perturbation of a layer of finite thickness is accurately represented by the partial derivative of the Haskell matrix of the layer,

$$\int_{z_{l-1}}^{z_l} \mathbf{a}(z_l - z)\mathbf{M}_{l,l}\mathbf{a}(z - z_{l-1})\mathrm{d}z = \mathbf{a}_{l,l}. \quad (35)$$

Dietrich & Kormendi (1990) derived analytic derivatives of the plane-wave response of a vertically heterogeneous medium based on eq. (31). They found that the accuracy deteriorates with

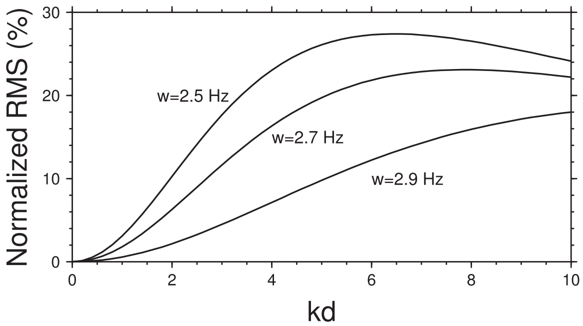

Figure 5. Relative differences between $\mathbf{a}_{,\beta}$ and numerical integration in eq. (35) under the thin-layer approximation as a function of layer thickness d (scaled by the wavenumber k of 1.0 s km^{-1}) at different frequencies. The layer shear-wave velocity $\beta = 3.0\,\mathrm{km\,s^{-1}}$ and density $\rho = 2.0\ \mathrm{g\ cm^{-3}}$.

increase in the perturbation layer thickness. Similar numerical tests by Moinet & Dietrich (1998) showed that the formulation was accurate only for incidence angles up to 35°. They attributed the cause to the Born approximation's limit on the layer thickness, which we now know not true. We believe that it is actually due to their thin-layer approximation or error in numerical integration in evaluating eq. (31). With the thin-layer approximation, eq. (31) becomes

$$\mathbf{a}_{l,l} \approx \mathbf{a}\left(\frac{d}{2}\right)\mathbf{M}_{l,l}\mathbf{a}\left(\frac{d}{2}\right), \tag{36}$$

where d is the layer thickness. We calculated the approximation error as a function of d and frequency (Fig. 5). As shown in the figure, the error increases with the layer thickness as well as the incidence angle (which is inversely proportional to the frequency).

5 CONCLUSIONS

In summary, we derived analytic partial derivatives of displacement kernels for both *P–SV* and *SH* waves generated by a point source in a multilayered half-space. They are expressed explicitly in terms of the analytic partial derivatives of the Haskell propagator matrix and the source vector. Differential seismograms are then computed using the wavenumber–frequency double integration method. We implemented an efficient algorithm to speed up the computation by storing intermediate matrix products. The total computation time for computing differential seismograms with respect to all the layers is only about one extra forward-problem computation time. Numerical tests show that our derivations are correct and the implementation is efficient. Both synthetic and real waveform data inversions using more accurate analytic differential seismograms perform better than using FD differential seismograms.

ACKNOWLEDGEMENTS

We thank Prof. Robert B. Herrmann for the helpful discussion. Comments from Prof. S. Mallick and an anonymous reviewer helped us to improve the manuscript greatly. This work was supported by NSF grant EAR-1249701 of USA.

REFERENCES

Berteussen, K.A., 1977. Moho depth determinations based on spectral-ratio analysis of NORSAR long-period P waves, *Phys. Earth planet. Inter.,* **15,** 13–27.

Bozdağ, E., Peter, D., Lefebvre, M., Komatitsch, D., Tromp, J., Hill, J., Podhorszki, N. & Pugmire, D., 2016. Global adjoint tomography: first-generation model, *Geophys. J. Int.,* **207,** 1739–1766.

Brandt, M.B.C., Grand, S.P., Nyblade, A.A. & Dirks, P.H.G.M., 2012. Upper mantle seismic structure beneath Southern Africa: constraints on the buoyancy supporting the African superswell, *Pure appl. Geophys.,* **169,** 595–614.

Chen, C., Gilbert, H., Andronicos, C., Hamburger, M.W., Larson, T., Marshak, S., Pavlis, G.L. & Yang, X., 2016. Shear velocity structure beneath the central United States: implications for the origin of the Illinois Basin and intraplate seismicity, *Geochem. Geophys. Geosyst.,* **17,** 1020–1041.

Chen, P., Zhao, L. & Jordan, T.H., 2007. Full 3D tomography for the crustal structure of the Los Angeles region, *Bull. seism. Soc. Am.,* **97,** 1094–1120.

Dietrich, M. & Kormendi, F., 1990. Perturbation of the plane-wave reflectivity of a depth-dependent elastic medium by weak inhomogeneities, *Geophys. J. Int.,* **100,** 203–214.

Du, Z.J., Michelini, A., Panza, G.F. & Urban, L., 1998. $P - SV$ multimode summation differential seismograms for layered structures, *Geophys. J. Int.,* **134,** 747–756.

Dunkin, J.W., 1965. Computation of modal solutions in layered elastic media at high frequencies, *Bull. seism. Soc. Am.,* **55,** 335–358.

Fichtner, A., Kennett, B., Igel, H. & Bunge, H., 2009. Full seismic waveform tomography for upper-mantle structure in the Australian region using adjoint methods, *Geophys. J. Int.,* **179,** 1703–1725.

Fuchs, K. & Müller, G., 1971. Computation of synthetic seismograms with the reflectivity method and comparison with observations, *Geophys. J. R. astr. Soc.,* **23,** 417–433.

Gao, W., Matzel, E. & Grand, S.P., 2006. Upper mantle seismic structure beneath eastern Mexico determined from *P* and *S* waveform inversion and its implications, *J. geophys. Res.,* **111,** B08307, doi:10.1029/2006JB004304.

Hadley, D. & Kanamori, H., 1977. Seismic structure of the Transverse Ranges, California, *Geol. soc. Am. Bull.,* **88,** 1469–1478.

Haskell, N.A., 1953. The dispersion of surface waves on multilayered media, *Bull. seism. Soc. Am.,* **43,** 17–34.

Haskell, N.A., 1963. Radiation pattern of Rayleigh waves from a fault of arbitrary dip and direction of motion in a homogeneous medium, *Bull. seism. Soc. Am.,* **51,** 495–502.

Herrmann, R.B., Withers, M. & Benz, H., 2008. The April 18, 2008 Illinois earthquake : An ANSS monitoring success, *Seismol. Res. Lett.,* **79,** 830–843.

Holt, W.E. & Wallace, T.C., 1990. Crustal thickness and upper mantle velocities in the Tibetan Plateau region from the inversion of regional *Pnl* waveforms: Evidence for a thick upper mantle Lid beneath Southern Tibet, *Geophys. Res. Lett.,* **12,** 12 499–12 525.

Hrubcová, P., Vavryčuk, V., Boušková, A. & Bohnhoff, M., 2016. Shallow crustal discontinuities inferred from waveforms of microearthquakes: Method and application to KTB Drill Site and West Bohemia Swarm Area, *J. geophys. Res.,* **121,** 881–902.

Hu, S. & Zhu, L., 2017. Calculation of differential seismograms using analytic partial derivatives – I. Teleseismic receiver functions, *Geophys. J. Int.,* **210**, 887–897.

Kennett, B., 1983. *Seismic Wave Propagation in Stratified Media,* Cambridge Univ. Press.

Kormendi, F. & Dietrich, M., 1991. Nonlinear waveform inversion of plane-wave seismograms in stratified elastic media, *Geophysics,* **56,** 664–674.

Liu, Q. & Gu, Y.J., 2012. Seismic imaging: From classical to adjoint tomography, *Tectonophysics,* **566–567,** 31–66.

Mallick, S. & Adhikari, S., 2015. Amplitude-variation-with-offset and prestack-waveform inversion: a direct comparison using a real data example from the Rock Spring Uplift, Wyoming, USA, *Geophysics,* **80**(2), B45–B59.

Moinet, F. & Dietrich, M., 1998. Computation of differential seismograms for point and plane scatterers in layered media, *Bull. seism. Soc. Am.,* **88,** 1311–1324.

Owens, T.J., Zandt, G. & Taylor, S.R., 1984. Seismic evidence for an ancient rift beneath the Cumberland plateau, Tennessee: A detailed analysis of broadband teleseismic *P* waveforms, *J. geophys. Res.,* **89,** 7783–7795.

Pafeng, J., Mallick, S. & Sharma, H., 2017. Prestack waveform inversion of three-dimensional seismic data—an example from the Rock Springs Uplift, Wyoming, USA, *Geophysics,* **82**(1), B1–B12.

Pan, G.S., Phinney, R.A. & Odom, R.I., 1988. Full-waveform inversion of plane-wave seismograms in stratified acoustic media: Theory and feasibility, *Geophysics,* **53,** 21–31.

Randall, G.E., 1994. Efficient calculation of complete differential seismograms for laterally homogeneous Earth models, *Geophys. J. Int.,* **109,** 323–342.

Sambridge, M.S. & Drijkoningen, G., 1992. Genetic algorithms in seismic waveform inversion, *Geophys. J. Int.,* **109,** 323–342.

Sen, M.K. & Roy, I.G., 2003. Computation of differential seismograms and iteration adaptive regularization in prestack waveform inversion, *Geophysics,* **68,** 2026–2039.

Takeuchi, H. & Saito, M., 1972. Seismic surface waves, in *Methods in Computational Physics,* vol. 11, pp. 217–295, ed. Bolt, B.A., Academic Press.

Tape, C., Liu, Q., Maggi, A. & Tromp, J., 2009. Adjoint tomography of the Southern California crust, *Science,* **325,** 988–992.

Tarantola, A., 1984. Inversion of seismic reflection data in the acoustic approximation, *Geophysics,* **49,** 1259–1266.

Virieux, J. & Operto, S., 2009. An overview of full-waveform inversion in exploration geophysics, *Geophysics,* **74**(6), WCC1–WCC26.

Wang, C.Y. & Herrmann, R.B., 1980. A numerical study of P, SV, and SH-wave generation in a plane layered medium, *Bull. seism. Soc. Am.,* **70,** 1015–1036.

Wang, Y., Grand, S.P. & Tang, Y., 2014. Shear velocity structure and mineralogy of the transition zone beneath the East Pacific Rise, *Earth planet. Sci. Lett.,* **402,** 313–323.

Watson, T., 1970. A note on fast computation of Rayleigh wave dispersion in the multilayered elastic half-space, *Bull. seism. Soc. Am.,* **60,** 161–166.

Xu, Y. & Wiens, D.A., 1997. Upper mantle structure of the southwest Pacific from regional waveform inversion, *J. geophys. Res.,* **102,** 27 439–27 451.

Yang, H., Zhu, L. & Chu, R., 2009. Fault plane determination of the April 18, 2008 Mt. Carmel, Illinois, earthquake by detecting and relocating aftershocks, *Bull. seism. Soc. Am.,* **99**(6), 3413–3420.

Zeng, Y. & Anderson, J.G., 1995. A method for direct computation of the differential seismogram with respect to the velocity change in a layered elastic solid, *Bull. seism. Soc. Am.,* **85,** 300–307.

Zhu, H., Bozdağ, E., Peter, D. & Tromp, J., 2012. Structure of the European upper mantle revealed by adjoint tomography, *Nat. Geosci.,* **5,** 493–498.

Zhu, L. & Rivera, L.A., 2002. A note on the dynamic and static displacements from a point source in multi-layered media, *Geophys. J. Int.,* **148,** 619–627.

APPENDIX A: PARTIAL DERIVATIVES FOR THE *P–SV* SYSTEM

The 6×6 *P–SV* compound matrix $\mathbf{a}|^{12,13,14,23,24,34}_{12,13,14,23,24,34}$ can be reduced to a 5×5 matrix $\mathscr{A}_{5\times5}$ by dropping off the third row and column of the original compound matrix and replacing its fourth row by $\left[2\mathbf{a}|^{23}_{12},\ 2\mathbf{a}|^{23}_{13},\ 2\mathbf{a}|^{23}_{23}-1,\ 2\mathbf{a}|^{23}_{24},\ 2\mathbf{a}|^{23}_{34}\right]$. Elements of $\mathscr{A}_{5\times5}$ are given in Watson (1970) and Wang & Herrmann (1980). The corresponding partial derivatives are

$$\frac{\partial \mathscr{A}_{11}}{\partial \alpha} = \frac{\gamma^2 k_\alpha^2}{\alpha k^2}\Big((1+\gamma_1^2)kdY_\alpha C_\beta - (kdC_\alpha + Y_\alpha)X_\beta - \gamma_1^2\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)Y_\beta\Big), \tag{A1}$$

$$\frac{\partial \mathscr{A}_{12}}{\partial \alpha} = \frac{\gamma k_\alpha^2}{2\alpha\mu k^2}\Big((kdC_\alpha + Y_\alpha)C_\beta - kdY_\alpha Y_\beta\Big), \tag{A2}$$

$$\frac{\partial \mathscr{A}_{13}}{\partial \alpha} = \frac{\gamma^2 k_\alpha^2}{2\alpha\mu k^2}\Big((1+\gamma_1)kdY_\alpha C_\beta - (kdC_\alpha + Y_\alpha)X_\beta - \gamma_1\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)Y_\beta\Big), \tag{A3}$$

$$\frac{\partial \mathscr{A}_{14}}{\partial \alpha} = \frac{\gamma k_\alpha^2}{2\alpha\mu k^2}\Big(\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)C_\beta - kdY_\alpha X_\beta\Big), \tag{A4}$$

$$\frac{\partial \mathscr{A}_{15}}{\partial \alpha} = \frac{\gamma^2 k_\alpha^2}{4\alpha\mu^2 k^2}\Big(2kdY_\alpha C_\beta - (kdC_\alpha + Y_\alpha)X_\beta - \frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)Y_\beta\Big), \tag{A5}$$

$$\frac{\partial \mathscr{A}_{21}}{\partial \alpha} = \frac{2\mu\gamma k_\alpha^2}{\alpha k^2}\Big(\gamma_1^2\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)C_\beta - kdY_\alpha X_\beta\Big), \tag{A6}$$

$$\frac{\partial \mathscr{A}_{22}}{\partial \alpha} = \frac{k_\alpha^2}{\alpha k^2}kdY_\alpha C_\beta, \tag{A7}$$

$$\frac{\partial \mathscr{A}_{23}}{\partial \alpha} = \frac{\gamma k_\alpha^2}{\alpha k^2}\Big(\gamma_1\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)C_\beta - kdY_\alpha X_\beta\Big), \tag{A8}$$

$$\frac{\partial \mathscr{A}_{24}}{\partial \alpha} = -\frac{k_\alpha^2}{\alpha\nu_\alpha^2}(kdC_\alpha - Y_\alpha)X_\beta, \tag{A9}$$

$$\frac{\partial \mathscr{A}_{25}}{\partial \alpha} = \frac{\partial \mathscr{A}_{14}}{\partial \alpha}, \tag{A10}$$

$$\frac{\partial \mathscr{A}_{31}}{\partial \alpha} = \frac{4\mu\gamma^2 k_\alpha^2}{\alpha k^2}\Big(\gamma_1^3\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)Y_\beta - (\gamma_1+\gamma_1^2)kdY_\alpha C_\beta + (kdC_\alpha + Y_\alpha)X_\beta\Big), \tag{A11}$$

$$\frac{\partial \mathscr{A}_{32}}{\partial \alpha} = \frac{2\gamma k_\alpha^2}{\alpha k^2}\Big(\gamma_1 kdY_\alpha Y_\beta - (kdC_\alpha + Y_\alpha)C_\beta\Big), \tag{A12}$$

$$\frac{\partial \mathscr{A}_{33}}{\partial \alpha} = \frac{2}{\alpha}\Big(\frac{k_\alpha^2}{k^2}kdY_\alpha C_\beta - \alpha\frac{\partial \mathscr{A}_{11}}{\partial \alpha}\Big), \tag{A13}$$

$$\frac{\partial \mathscr{A}_{34}}{\partial \alpha} = -2\frac{\partial \mathscr{A}_{23}}{\partial \alpha}, \tag{A14}$$

$$\frac{\partial \mathscr{A}_{35}}{\partial \alpha} = -2\frac{\partial \mathscr{A}_{13}}{\partial \alpha}, \tag{A15}$$

$$\frac{\partial \mathscr{A}_{41}}{\partial \alpha} = \frac{2\mu\gamma k_\alpha^2}{\alpha k^2}\Big((kdC_\alpha + Y_\alpha)C_\beta - \gamma_1^2 kdY_\alpha Y_\beta\Big), \tag{A16}$$

$$\frac{\partial \mathscr{A}_{42}}{\partial \alpha} = -\frac{k_\alpha^2}{\alpha k^2}(kdC_\alpha + Y_\alpha)Y_\beta, \tag{A17}$$

$$\frac{\partial \mathscr{A}_{43}}{\partial \alpha} = -\frac{1}{2}\frac{\partial \mathscr{A}_{32}}{\partial \alpha}, \tag{A18}$$

$$\frac{\partial \mathscr{A}_{44}}{\partial \alpha} = \frac{\partial \mathscr{A}_{22}}{\partial \alpha}, \tag{A19}$$

$$\frac{\partial \mathscr{A}_{45}}{\partial \alpha} = \frac{\partial \mathscr{A}_{12}}{\partial \alpha}, \tag{A20}$$

$$\frac{\partial \mathscr{A}_{51}}{\partial \alpha} = \frac{4\mu^2\gamma^2 k_\alpha^2}{\alpha k^2}\Big(2kd\gamma_1^2 Y_\alpha C_\beta - (kdC_\alpha + Y_\alpha)X_\beta - \gamma_1^4\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)Y_\beta\Big), \tag{A21}$$

$$\frac{\partial \mathscr{A}_{52}}{\partial \alpha} = \frac{\partial \mathscr{A}_{41}}{\partial \alpha}, \tag{A22}$$

$$\frac{\partial \mathscr{A}_{53}}{\partial \alpha} = -\frac{1}{2}\frac{\partial \mathscr{A}_{31}}{\partial \alpha}, \tag{A23}$$

$$\frac{\partial \mathscr{A}_{54}}{\partial \alpha} = \frac{\partial \mathscr{A}_{21}}{\partial \alpha}, \tag{A24}$$

$$\frac{\partial \mathscr{A}_{55}}{\partial \alpha} = \frac{\partial \mathscr{A}_{11}}{\partial \alpha}, \tag{A25}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{11}}{\partial \beta} = \frac{2\gamma^2}{\beta}\Big(&2(1+\gamma_1)C_\alpha C_\beta + \frac{2}{\gamma}(\gamma_1 + \frac{1}{2\gamma^2})kdC_\alpha Y_\beta \\ &- 2X_\alpha X_\beta - \frac{1}{\gamma}X_\alpha(kdC_\beta + Y_\beta) - 2\gamma_1 Y_\alpha Y_\beta \\ &- \gamma_1^2\gamma_3 Y_\alpha(kdC_\beta - Y_\beta) - 2(1+\gamma_1)\Big),\end{aligned} \tag{A26}$$

$$\frac{\partial \mathscr{A}_{12}}{\partial \beta} = \frac{1}{\beta\mu}\Big(kdX_\alpha Y_\beta - \frac{k^2}{\nu_\beta^2}C_\alpha(kdC_\beta - Y_\beta)\Big), \tag{A27}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{13}}{\partial \beta} = \frac{\gamma^2}{2\beta\mu}\Big(&4(C_\alpha C_\beta - 1) - 2X_\alpha X_\beta - 2Y_\alpha Y_\beta \\ &+ 2\frac{1+\gamma_1}{\gamma}kdC_\alpha Y_\beta - \frac{2}{\gamma}X_\alpha(kdC_\beta + Y_\beta) \\ &- 2\gamma_1\gamma_3(kdC_\beta - Y_\beta)Y_\alpha\Big),\end{aligned} \tag{A28}$$

$$\frac{\partial \mathscr{A}_{14}}{\partial \beta} = \frac{1}{\beta\mu}\Big(kdY_\alpha Y_\beta - C_\alpha(kdC_\beta + Y_\beta)\Big), \tag{A29}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{15}}{\partial \beta} = \frac{\gamma}{2\beta\mu^2}\Big(&2kdC_\alpha Y_\beta - X_\alpha(kdC_\beta + Y_\beta) \\ &- \gamma\gamma_3 Y_\alpha(kdC_\beta - Y_\beta)\Big),\end{aligned} \tag{A30}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{21}}{\partial \beta} = \frac{4\mu\gamma}{\beta}\Big(&2\gamma_1 Y_\alpha C_\beta - 2C_\alpha X_\beta + \frac{\gamma_1^2}{\gamma}kdY_\alpha Y_\beta \\ &- \frac{1}{\gamma}C_\alpha(kdC_\beta + Y_\beta)\Big),\end{aligned} \tag{A31}$$

$$\frac{\partial \mathscr{A}_{22}}{\partial \beta} = \frac{2kd}{\beta\gamma}C_\alpha Y_\beta, \tag{A32}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{23}}{\partial \beta} = \frac{2}{\beta}\Big(&\gamma Y_\alpha C_\beta - \gamma C_\alpha X_\beta - \gamma_1 kdY_\alpha Y_\beta \\ &- C_\alpha(kdC_\beta + Y_\beta)\Big),\end{aligned} \tag{A33}$$

$$\frac{\partial \mathscr{A}_{24}}{\partial \beta} = -\frac{2}{\beta\gamma}Y_\alpha(kdC_\beta + Y_\beta), \tag{A34}$$

$$\frac{\partial \mathscr{A}_{25}}{\partial \beta} = \frac{\partial \mathscr{A}_{14}}{\partial \beta}, \tag{A35}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{31}}{\partial \beta} = \frac{8\mu\gamma^2}{\beta}\Big(&3\gamma_1^2 Y_\alpha Y_\beta + \gamma_1^3\gamma_3 Y_\alpha(kdC_\beta - Y_\beta) \\ &- (6\gamma_1 + \frac{1}{\gamma^2})(C_\alpha C_\beta - 1) \\ &- \frac{\gamma_1(1+\gamma_1)}{\gamma}kdC_\alpha Y_\beta + 3X_\alpha X_\beta \\ &+ \frac{1}{\gamma}X_\alpha(kdC_\beta + Y_\beta)\Big),\end{aligned} \tag{A36}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{32}}{\partial \beta} = \frac{4\gamma}{\beta}\Big(&C_\alpha Y_\beta - X_\alpha C_\beta + \gamma_1\gamma_3 C_\alpha(kdC_\beta - Y_\beta) \\ &- \frac{1}{\gamma}kdX_\alpha Y_\beta\Big),\end{aligned} \tag{A37}$$

$$\frac{\partial \mathscr{A}_{33}}{\partial \beta} = \frac{2}{\beta\gamma}\Big(kdC_\alpha Y_\beta - \beta\gamma\frac{\partial \mathscr{A}_{11}}{\partial \beta}\Big), \tag{A38}$$

$$\frac{\partial \mathscr{A}_{34}}{\partial \beta} = -2\frac{\partial \mathscr{A}_{23}}{\partial \beta}, \tag{A39}$$

$$\frac{\partial \mathscr{A}_{35}}{\partial \beta} = -2\frac{\partial \mathscr{A}_{13}}{\partial \beta}, \tag{A40}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{41}}{\partial \beta} = \frac{4\mu\gamma}{\beta}\Big(&2X_\alpha C_\beta - 2\gamma_1 C_\alpha Y_\beta + \frac{1}{\gamma}kdX_\alpha Y_\beta \\ &- \gamma_1^2\gamma_3 C_\alpha(kdC_\beta - Y_\beta)\Big),\end{aligned} \tag{A41}$$

$$\frac{\partial \mathscr{A}_{42}}{\partial \beta} = -\frac{2\gamma_3}{\beta}X_\alpha(kdC_\beta - Y_\beta), \tag{A42}$$

$$\frac{\partial \mathscr{A}_{43}}{\partial \beta} = -\frac{1}{2}\frac{\partial \mathscr{A}_{32}}{\partial \beta}, \tag{A43}$$

$$\frac{\partial \mathscr{A}_{44}}{\partial \beta} = \frac{\partial \mathscr{A}_{22}}{\partial \beta}, \tag{A44}$$

$$\frac{\partial \mathscr{A}_{45}}{\partial \beta} = \frac{\partial \mathscr{A}_{12}}{\partial \beta}, \tag{A45}$$

$$\begin{aligned}\frac{\partial \mathscr{A}_{51}}{\partial \beta} = \frac{8\mu^2\gamma^2}{\beta}\Big(&4\gamma_1(\gamma_1+1)(C_\alpha C_\beta - 1) - 4X_\alpha X_\beta \\ &- 4\gamma_1^3 Y_\alpha Y_\beta + \frac{2\gamma_1^2}{\gamma}kdC_\alpha Y_\beta \\ &- \frac{1}{\gamma}X_\alpha(kdC_\beta + Y_\beta) \\ &- \gamma_1^4\gamma_3 Y_\alpha(kdC_\beta - Y_\beta)\Big),\end{aligned} \tag{A46}$$

$$\frac{\partial \mathscr{A}_{52}}{\partial \beta} = \frac{\partial \mathscr{A}_{41}}{\partial \beta}, \tag{A47}$$

$$\frac{\partial \mathscr{A}_{53}}{\partial \beta} = -\frac{1}{2}\frac{\partial \mathscr{A}_{31}}{\partial \beta}, \tag{A48}$$

$$\frac{\partial \mathscr{A}_{54}}{\partial \beta} = \frac{\partial \mathscr{A}_{21}}{\partial \beta}, \tag{A49}$$

$$\frac{\partial \mathscr{A}_{55}}{\partial \beta} = \frac{\partial \mathscr{A}_{11}}{\partial \beta}, \tag{A50}$$

$$\frac{\partial \mathscr{A}_{12}}{\partial \rho} = -\frac{\gamma}{2\rho\mu}(X_\alpha C_\beta - C_\alpha Y_\beta), \tag{A51}$$

$$\frac{\partial \mathscr{A}_{13}}{\partial \rho} = -\frac{\gamma^2}{2\rho\mu}\Big((1+\gamma_1)(C_\alpha C_\beta - 1) - X_\alpha X_\beta - \gamma_1 Y_\alpha Y_\beta\Big), \tag{A52}$$

$$\frac{\partial \mathscr{A}_{14}}{\partial \rho} = -\frac{\gamma}{2\rho\mu}(Y_\alpha C_\beta - C_\alpha X_\beta), \tag{A53}$$

$$\frac{\partial \mathscr{A}_{15}}{\partial \rho} = -\frac{\gamma^2}{2\rho\mu^2}\Big(2(C_\alpha C_\beta - 1) - X_\alpha X_\beta - Y_\alpha Y_\beta\Big), \tag{A54}$$

$$\frac{\partial \mathscr{A}_{21}}{\partial \rho} = \frac{2\gamma\mu}{\rho}(\gamma_1^2 Y_\alpha C_\beta - C_\alpha X_\beta), \tag{A55}$$

$$\frac{\partial \mathscr{A}_{25}}{\partial \rho} = \frac{\partial \mathscr{A}_{14}}{\partial \rho}, \tag{A56}$$

$$\frac{\partial \mathscr{A}_{31}}{\partial \rho} = \frac{4\mu\gamma^2}{\rho}\Big(\gamma_1^3 Y_\alpha Y_\beta - \gamma_1(1+\gamma_1)(C_\alpha C_\beta - 1) + X_\alpha X_\beta\Big), \tag{A57}$$

$$\frac{\partial \mathscr{A}_{35}}{\partial \rho} = -2\frac{\partial \mathscr{A}_{13}}{\partial \rho}, \tag{A58}$$

$$\frac{\partial \mathscr{A}_{41}}{\partial \rho} = \frac{2\mu\gamma}{\rho}(X_\alpha C_\beta - \gamma_1^2 C_\alpha Y_\beta), \tag{A59}$$

$$\frac{\partial \mathscr{A}_{45}}{\partial \rho} = \frac{\partial \mathscr{A}_{12}}{\partial \rho}, \tag{A60}$$

$$\frac{\partial \mathscr{A}_{51}}{\partial \rho} = \frac{8\mu^2\gamma^2}{\rho}\Big(2(C_\alpha C_\beta - 1)\gamma_1^2 - X_\alpha X_\beta - \gamma_1^4 Y_\alpha Y_\beta\Big), \tag{A61}$$

$$\frac{\partial \mathscr{A}_{52}}{\partial \rho} = \frac{\partial \mathscr{A}_{41}}{\partial \rho}, \tag{A62}$$

$$\frac{\partial \mathscr{A}_{53}}{\partial \rho} = -\frac{1}{2}\frac{\partial \mathscr{A}_{31}}{\partial \rho}, \tag{A63}$$

$$\frac{\partial \mathscr{A}_{54}}{\partial \rho} = \frac{\partial \mathscr{A}_{21}}{\partial \rho}, \tag{A64}$$

$$\frac{\partial \mathscr{A}_{55}}{\partial \rho} = \frac{\partial \mathscr{A}_{11}}{\partial \rho}. \tag{A65}$$

The 1×5 vector $\mathbf{g} = \frac{8\mu\nu_\beta\nu_\alpha}{\gamma k^2}\left[\mathbf{E}^{-1}|^{12}_{12,13,23,24,34}\right]$ is given in Zhu & Rivera (2002). Its partial derivatives are

$$\frac{\partial \mathbf{g}}{\partial \alpha} = \frac{1}{\alpha}\frac{k_\alpha^2}{k^2\nu_\alpha}\left[-2\mu\gamma\nu_\beta,\ k,\ -\gamma\nu_\beta,\ 0,\ -\frac{\gamma\nu_\beta}{2\mu}\right], \tag{A66}$$

$$\frac{\partial \mathbf{g}}{\partial \beta} = \frac{1}{\beta}\left[4\mu(2\delta - \frac{\nu_\alpha}{\nu_\beta}),\ 0,\ 2\left(\delta + 1 - \frac{\nu_\alpha}{\nu_\beta}\right),\ \frac{2k}{\gamma\nu_\beta},\ -\frac{\nu_\alpha}{\mu\nu_\beta}\right], \tag{A67}$$

$$\frac{\partial \mathbf{g}}{\partial \beta} = \frac{1}{\rho}\left[2\mu(\delta - \gamma_1),\ 0,\ 0,\ 0,\ -\frac{\delta+1}{2\mu}\right], \tag{A68}$$

where $\delta = \gamma(1 - \nu_\alpha\nu_\beta/k^2) - 1$.

The partial derivatives of the explosion source vector are,

$$\frac{\partial \mathbf{s}^0}{\partial \alpha} = \frac{1}{\alpha}\left[0,\ -\frac{2}{\rho\alpha^2},\ 0,\ -\frac{4\beta^2}{\alpha^2}\right]^{\mathrm{T}}, \tag{A69}$$

$$\frac{\partial \mathbf{s}^0}{\partial \beta} = \frac{1}{\beta}\left[0,\ 0,\ 0,\ \frac{4\beta^2}{\alpha^2},\ 0,\ 0\right]^{\mathrm{T}}, \tag{A70}$$

$$\frac{\partial \mathbf{s}^0}{\partial \rho} = \frac{1}{\rho}\left[0,\ -\frac{1}{\rho\alpha^2},\ 0,\ 0\right]^{\mathrm{T}}. \tag{A71}$$

The partial derivatives of the single-force source vector are all zero. The non-zero partial derivatives of the double-couple source vector are

$$\frac{\partial \mathbf{s}^0}{\partial \alpha} = \frac{1}{\alpha}\left[0,\ -\frac{4}{\rho\alpha^2},\ 0,\ -\frac{8\beta^2}{\alpha^2}\right]^{\mathrm{T}}, \tag{A72}$$

$$\frac{\partial \mathbf{s}^0}{\partial \beta} = \frac{1}{\beta}\left[0,\ 0,\ 0,\ \frac{8\beta^2}{\alpha^2}\right]^{\mathrm{T}}, \tag{A73}$$

$$\frac{\partial \mathbf{s}^1}{\partial \beta} = \frac{1}{\beta}\left[-\frac{2}{\rho\beta^2},\ 0,\ 0,\ 0\right]^{\mathrm{T}}, \tag{A74}$$

$$\frac{\partial \mathbf{s}^0}{\partial \rho} = \frac{1}{\rho}\left[0,\ -\frac{2}{\rho\alpha^2},\ 0,\ 0\right]^{\mathrm{T}}, \tag{A75}$$

$$\frac{\partial \mathbf{s}^1}{\partial \rho} = \frac{1}{\rho}\left[-\frac{1}{\rho\beta^2},\ 0,\ 0,\ 0\right]^{\mathrm{T}}. \tag{A76}$$

APPENDIX B: PARTIAL DERIVATIVES FOR THE *SH* SYSTEM

The non-zero partial derivatives of the *SH* Haskell matrix are

$$\frac{\partial a_{11}}{\partial \beta} = \frac{2kdY_\beta}{\beta\gamma}, \tag{B1}$$

$$\frac{\partial a_{12}}{\partial \beta} = -\frac{2}{\beta\mu}\big(\gamma_3(kdC_\beta - Y_\beta) - Y_\beta\big), \tag{B2}$$

$$\frac{\partial a_{21}}{\partial \beta} = -\frac{2\mu}{\beta\gamma}\big(\gamma X_\beta + kdC_\beta + Y_\beta\big), \tag{B3}$$

$$\frac{\partial a_{22}}{\partial \beta} = \frac{\partial a_{11}}{\partial \beta}, \tag{B4}$$

$$\frac{\partial a_{12}}{\partial \rho} = \frac{1}{\rho\mu}Y_\beta, \tag{B5}$$

$$\frac{\partial a_{21}}{\partial \rho} = -\frac{1}{\rho}\mu X_\beta, \tag{B6}$$

$$\frac{\partial \mathbf{E}^{-1}}{\partial \beta} = -\frac{k}{\beta\mu}\begin{bmatrix} 0 & \frac{1+\gamma_3}{\nu_\beta} \\ 0 & \frac{1+\gamma_3}{\nu_\beta} \end{bmatrix}, \tag{B7}$$

$$\frac{\partial \mathbf{E}^{-1}}{\partial \rho} = \frac{1}{2\rho}\begin{bmatrix} 0 & -\frac{k}{\mu\nu_\beta} \\ 0 & -\frac{k}{\mu\nu_\beta} \end{bmatrix}. \tag{B8}$$

The non-zero partial derivatives of the *SH* double-couple source terms are,

$$\frac{\partial s_1^1}{\partial \beta} = \frac{2}{\rho\beta^3}, \tag{B9}$$

$$\frac{\partial s_1^1}{\partial \rho} = \frac{2}{\rho^2\beta^2}. \tag{B10}$$

APPENDIX C: VIRTUAL SOURCE VECTORS OF THE DIFFERENTIAL WAVEFIELD

By taking partial derivative to eq. (1) with respect to a medium parameter in layer l and assuming that the source is not in this layer, we have

$$\frac{\mathrm{d}\mathbf{b}_{,l}(z)}{\mathrm{d}z} = \mathbf{M}_l\mathbf{b}_{,l}(z) + \mathbf{M}_{l,l}\mathbf{b}(z),\ \text{if} z_l \le z \le z_{l-1}, \tag{C1}$$

$$\frac{\mathrm{d}\mathbf{b}_{,l}(z)}{\mathrm{d}z} = \mathbf{M}_i \mathbf{b}_{,l}(z), \text{ otherwise.} \tag{C2}$$

Following Takeuchi & Saito (1972) and Kennett (1983), we integrate eq. (C1) from the top to the bottom of the layer and get,

$$\mathbf{b}_{,l}(z_l) = \mathbf{a}_l \mathbf{b}_{,l}(z_{l-1}) + \int_{z_{-1}}^{z_l} \mathbf{a}(z_l - z)\mathbf{M}_{l,l}\mathbf{b}(z)\mathrm{d}z. \tag{C3}$$

So the virtual source vector, that is, displacement-stress jump to $\mathbf{b}_{,l}$ across the layer is

$$\begin{aligned}
\mathbf{s}^{\mathrm{v}} &= \mathbf{a}_l \mathbf{b}_{,l}(z_{l-1}) - \mathbf{b}_{,l}(z_l) \\
&= -\int_{z_{-1}}^{z_l} \mathbf{a}(z_l - z)\mathbf{M}_{l,l}\mathbf{b}(z)\mathrm{d}z \\
&= -\int_{z_{l-1}}^{z_l} \mathbf{a}(z_l - z)\mathbf{M}_{l,l}\mathbf{a}(z - z_{l-1})\mathrm{d}z\mathbf{b}(z_{l-1}),
\end{aligned} \tag{C4}$$

as in eq. (31)

Next, we take partial derivative to eq. (2) with respect to medium parameter in layer l,

$$\mathbf{b}_{,l}(z) = \mathbf{E\Lambda}(z)\mathbf{w}_{,l} + \left(\mathbf{E\Lambda}(z)\right)_{,l}\mathbf{w}. \tag{C5}$$

the virtual source vector

$$\begin{aligned}
\mathbf{s}^{\mathrm{v}} &= \mathbf{a}_l \mathbf{b}_{,l}(z_{l-1}) - \mathbf{b}_{,l}(z_l) \\
&= \mathbf{a}_l \mathbf{E\Lambda}(z_{l-1})\mathbf{w}_{,l} + \mathbf{a}_l\left(\mathbf{E\Lambda}(z_{l-1})\right)_{,l}\mathbf{w} \\
&\quad - \mathbf{E\Lambda}(z_l)\mathbf{w}_{,l} - \left(\mathbf{E\Lambda}(z_l)\right)_{,l}\mathbf{w} \\
&= \mathbf{a}_l\left(\mathbf{E\Lambda}(z_{l-1})\right)_{,l}\mathbf{w} - \left(\mathbf{E\Lambda}(z_l)\right)_{,l}\mathbf{w} \\
&= \left(\mathbf{a}_l\mathbf{E\Lambda}(z_{l-1}) - \mathbf{E\Lambda}(z_l)\right)_{,l}\mathbf{w} - \mathbf{a}_{l,l}\mathbf{E\Lambda}(z_{l-1})\mathbf{w} \\
&= -\mathbf{a}_{l,l}\mathbf{b}_{l-1},
\end{aligned} \tag{C6}$$

as in eq. (32). In the above derivation, we used

$$\mathbf{E}^{-1}\mathbf{E}_{,l} = -\mathbf{E}_{,l}^{-1}\mathbf{E}, \tag{C7}$$

$$\mathbf{\Lambda}^{-1}\mathbf{\Lambda}_{,l} = -\mathbf{\Lambda}_{,l}^{-1}\mathbf{\Lambda}, \tag{C8}$$

$$\mathbf{w} = \mathbf{\Lambda}^{-1}(z_{l-1})\mathbf{E}^{-1}\mathbf{b}(z_{l-1}). \tag{C9}$$

If layer l is the source layer, that is, $l = m$ or $m + 1$, the wave amplitude vector in eq. (C5) is z-dependent:

$$\mathbf{w}(z) = \begin{cases} \mathbf{w}_m, & \text{if } z_{m-1} > z > z_m, \\ \mathbf{w}_{m+1}, & \text{if } z_m > z > z_{m+1}, \end{cases} \tag{C10}$$

$$\mathbf{E\Lambda}(z_m)\left(\mathbf{w}_m - \mathbf{w}_{m+1}\right) = \mathbf{s}. \tag{C11}$$

The virtual source vector is,

$$\begin{aligned}
\mathbf{s}^{\mathrm{v}} &= \mathbf{a}_{m+1}\mathbf{a}_m\mathbf{b}_{,l}(z_{m-1}) - \mathbf{b}_{,l}(z_{m+1}) \\
&= \mathbf{a}_{m+1}\mathbf{a}_m\left(\mathbf{E\Lambda}(z_{m-1})\mathbf{w}_{m,l} + \left(\mathbf{E\Lambda}(z_{m-1})\right)_{,l}\mathbf{w}_m\right) \\
&\quad - \mathbf{E\Lambda}(z_{m+1})\mathbf{w}_{m+1,l} - \left(\mathbf{E\Lambda}(z_{m+1})\right)_{,l}\mathbf{w}_{m+1} \\
&= \mathbf{E\Lambda}(z_{m+1})\mathbf{w}_{m,l} + \mathbf{a}_{m+1}\mathbf{a}_m\left(\mathbf{E\Lambda}(z_{m-1})\right)_{,l}\mathbf{w}_m \\
&\quad - \mathbf{E\Lambda}(z_{m+1})\mathbf{w}_{m+1,l} - \left(\mathbf{E\Lambda}(z_{m+1})\right)_{,l}\mathbf{w}_{m+1} \\
&= \mathbf{E\Lambda}(z_{m+1})(\mathbf{w}_m - \mathbf{w}_{m+1})_{,l} + \mathbf{a}_{m+1}\mathbf{a}_m\left(\mathbf{E\Lambda}(z_{m-1})\right)_{,l}\mathbf{w}_m \\
&\quad - \left(\mathbf{a}_{m+1}\mathbf{a}_m\mathbf{E\Lambda}(z_{m-1})\right)_{,l}\mathbf{w}_{m+1} \\
&= \mathbf{a}_{m+1}\mathbf{a}_m\mathbf{E\Lambda}(z_{m-1})(\mathbf{w}_m - \mathbf{w}_{m+1})_{,l} \\
&\quad + \mathbf{a}_{m+1}\mathbf{a}_m\left(\mathbf{E\Lambda}(z_{m-1})\right)_{,l}(\mathbf{w}_m - \mathbf{w}_{m+1}) \\
&\quad - (\mathbf{a}_{m+1}\mathbf{a}_m)_{,l}\mathbf{E\Lambda}(z_{m-1})\mathbf{w}_{m+1} \\
&= \mathbf{a}_{m+1}\mathbf{a}_m\left(\mathbf{E\Lambda}(z_{m-1})(\mathbf{w}_m - \mathbf{w}_{m+1})\right)_{,l} \\
&\quad - (\mathbf{a}_{m+1}\mathbf{a}_m)_{,l}\mathbf{E\Lambda}(z_{m-1})\mathbf{w}_{m+1} \\
&= \left(\mathbf{a}_{m+1}\mathbf{a}_m\mathbf{E\Lambda}(z_{m-1})(\mathbf{w}_m - \mathbf{w}_{m+1})\right)_{,l} \\
&\quad - (\mathbf{a}_{m+1}\mathbf{a}_m)_{,l}\mathbf{E\Lambda}(z_{m-1})\mathbf{w}_m \\
&= \left(\mathbf{a}_{m+1}\mathbf{E\Lambda}(z_m)(\mathbf{w}_m - \mathbf{w}_{m+1})\right)_{,l} - (\mathbf{a}_{m+1}\mathbf{a}_m)_{,l}\mathbf{b}_{m-1} \\
&= (\mathbf{a}_{m+1}\mathbf{s})_{,l} - (\mathbf{a}_{m+1}\mathbf{a}_m)_{,l}\mathbf{b}_{m-1} \\
&= \mathbf{a}_{m+1,l}\mathbf{s} + \mathbf{a}_{m+1}\mathbf{s}_{,l} - (\mathbf{a}_{m+1}\mathbf{a}_m)_{,l}\mathbf{b}_{m-1}.
\end{aligned} \tag{C12}$$